Date: .....

Assignment No: 1

Topic :
Number conversition

Subject:
computer orgnizition

Submitted by:
Safwan Ahmad

Submitted to:
Sir Zeeshan

Reg-No:
AI120252038

Department:
AI

Ichhat university of science and technology (ICUST).

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Number conversition:-

1) Binary to decimal:-

Question:-

$(1011001101)_2 = (?)_{10}$

Solution:-

(1011001101)

→ $1\times2^9+0\times2^8+1\times2^7+1\times2^6+0\times2^5+0\times2^4+1\times2^3+1\times2^2+0\times2^1+1\times2^0$

→ $1\times512+0\times256+1\times128+1\times64+0\times32+0\times16+1\times8+1\times4+0\times2+1\times1$

→ $512+0+128+64+0+0+8+4+0+1$

→ $512+128+64+8+4+1$

→ 717

Answer $(1011001101)_2 = (717)_{10}$

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2) Decimal to Binary:

Question:

$(347)_{10} = (?)_2$

Solution:

$(347)_{10} = (?)_2$

Step 1: We divid 347 by 2

2	347	1
2	173	1
2	86	0
2	43	1
2	21	1
2	10	0
2	5	1
2	2	0
	1	

Step 2: We write reminder from bottom to top.

Answer: $(347)_{10} = (101011011)_2$

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3) Octal to decimal:

Question:

$$(5726)_8 = (?)_{10}$$

Solution:

$$(5726)_8 = (?)_{10}$$

$$\rightarrow 5 \times 8^3 + 7 \times 8^2 + 2 \times 8^1 + 6 \times 8^0$$

$$\rightarrow 5 \times 512 + 7 \times 64 + 2 \times 8 + 6 \times 1$$

$$\rightarrow 2560 + 448 + 16 + 6$$

$$\rightarrow 3030$$

Answer: $(5726)_8 = (3030)_{10}$

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4) Decimal to octal

Question:

$(5839)_{10} = (?)_8$

Solution:

$(5839)_{10} = (?)_8$

Step 1: First we divided 5839 by 8.

8	5839	7
8	729	1
8	91	3
8	11	3
8	1	1
	0	

step 2: write the reminder from bottom to top

(13317)

Answer: $(5839)_{10} = (13317)_8$

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5) Hexadecimal to octal

Question:

$(3A7)_{16} = (?)_8$

Solution:

$(3A7)_{16} = (?)_8$

Step (1) we convert them into binary

→ 3 = 0011
→ A = 1010
→ 7 = 0111

So

$(3A7)_{16} = (001110100111)_2$

Step (2) we convert them into octal

Group the binary digits into groups of 3 from right to left

(001 110 100 111)

001 110 100 111

Convert each group

→ 001 = 1
→ 110 = 6 → (1647)
→ 100 = 4
→ 111 = 7

Answer: $(3A7)_{16} = (1647)_8$

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6) Octal to Hexadecimal :

Question

$(5726)_8 = (?)_{16}$

Solution:

$(5726)_8 = (?)_{16}$

Step 1 Convert each octal digit into 3-bit binary

→ 5 = 101
→ 7 = 111
→ 2 = 010
→ 6 = 110

$(5726)_8 = (101111010110)_2$

Step 2: Group the binary digit into groups of 4 from right to left.

(101111010110)

1011 1101 0110

Convert each group

→ 1011 = B
→ 1101 = D
→ 0110 = 6

Answer: $(5726)_8 = (BD6)_{16}$

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